

ACADEMIC DISSERTATION • MASTER OF SCIENCE RESEARCH
THESIS

FAIRMETACHAIN: A BLOCKCHAIN-BASED FAIR RESOURCE ALLOCATION FRAMEWORK FOR METAVERSE APPLICATIONS

*A Game-Theoretic, Cryptographically Verified, and Gas-Efficient Infrastructure
for Scalable Decentralized Spatial Computing*

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Comparative Baseline: IEEE VTC-2022 MetaChain Architecture

Smart Contract Generation: FairMetaChain V6 (Production Edition)

ABSTRACT

The rapid evolution of persistent synthetic realities and Metaverse environments has catalyzed unprecedented demand for scalable, low-latency, and decentralized edge computing infrastructure. To coordinate heterogeneous computational resources across decentralized Metaverse shards, existing literature—most notably the foundational MetaChain framework (IEEE VTC-2022)—employs a two-stage Stackelberg game with a pure pay-per-share reward mechanism. However, comprehensive empirical and game-theoretic analysis reveals four critical structural failures in the baseline architecture: (1) **Whale Monopolization**, where high-capacity nodes capture upwards of 90% of the epoch rewards, starving resource-constrained edge contributors; (2) the **Fairness-Sybil Dilemma**, wherein naive fairness caps inadvertently reward identity-splitting attackers with up to $1.75\times$ higher profit; (3) **Uncontrolled Transaction Congestion**, arising from the complete absence of dynamic gas economics and progressive rebates; and (4) **Vulnerability to Compute Spoofing**, caused by reliance on unverified, self-reported resource claims.

To address and overcome these fundamental limitations, this dissertation designs, implements, and evaluates **FairMetaChain**, a unified game-theoretic and cryptographically verified resource allocation framework. FairMetaChain introduces a 35% fair-share allocation ceiling modulated by continuous Exponential Moving Average (EMA) reputation dynamics ($\phi \in [0.2, 1.0]$), systematically elevating edge node viability. To resolve the Fairness-Sybil Dilemma, we construct a *Dual-Mode Sybil Neutrality* defense integrating Trusted Execution Environment (TEE) hardware remote attestations (Intel SGX / AMD SEV), non-refundable recycled entrance fees (5 FMC), and reputation warm-up haircuts. We theoretically prove and empirically demonstrate that identity splitting yields a strictly negative net utility (-1.20 FMC vs. $+8.34$ FMC for honest nodes; a net penalty of -9.54 FMC). Furthermore, FairMetaChain establishes a dynamic per-shard gas model coupled with a 40% progressive gas rebate pool distributed strictly proportional to gas paid, rendering 1-wei dust transaction exploits completely unprofitable.

The framework is fully implemented across six iterative smart contract generations (V1 to V6) written in Solidity 0.8.20, featuring EVM storage slot packing (28.2% gas savings), multi-epoch atomic batch claims (39.2% gas savings), off-chain cryptographic work receipts (`contributeWithProof`), and autonomous Chainlink-compatible keeper orchestration. Rigorous Hardhat EVM benchmarks across 11 heterogeneous node classes validate that FairMetaChain improves Jain's Fairness Index by **31.4%** (from 0.648 to 0.852), eliminates compute spoofing, and achieves trustless, operator-free epoch progression.

Keywords: Metaverse Computing, Blockchain Sharding, Resource Allocation, Stackelberg Game, Sybil Resistance, Trusted Execution Environments, Smart Contract Optimization, Gas Economics, Jain's Fairness Index.

FAIRMETACHAIN: A BLOCKCHAIN-BASED FAIR RESOURCE ALLOCATION FRAMEWORK FOR METAVERSE APPLICATIONS

A Thesis Submitted in Partial Fulfillment of the Requirements for the Degree of Master of Science in Computer Science & Engineering

ABSTRACT

The rapid evolution of the Metaverse has catalyzed massive demand for scalable, low-latency, and decentralized computing infrastructure capable of rendering complex 3D virtual environments, executing spatial physics simulations, and facilitating trustless digital asset exchanges. While blockchain sharding offers a promising foundation for cross-application scalability, existing resource allocation frameworks—such as the baseline MetaChain model (IEEE VTC-2022)—rely on pure pay-per-share incentive mechanisms. This creates severe structural failures in decentralized environments: (1) high-capacity "whale" nodes monopolize over 90% of network rewards, starving resource-constrained edge devices; (2) naive fairness caps introduce the "Fairness–Sybil Dilemma," enabling whales to split identities into multiple accounts and capture up to $1.75\times$ more profit; (3) transaction congestion is left unmitigated due to the absence of gas economic dynamics; and (4) compute contributions are assumed to be honest without cryptographic verification.

To overcome these foundational limitations, this thesis proposes **FairMetaChain**, a comprehensive, game-theoretically grounded, and cryptographically verified resource allocation framework. FairMetaChain integrates a 35% fair-share cap with an Exponential Moving Average (EMA) reputation multiplier ($\varphi \in [0.2, 1.0]$), guaranteeing equitable compensation for edge contributors. To resolve the Fairness–Sybil Dilemma, the framework introduces **Dual-Mode Sybil Neutrality** combined with Trusted Execution Environment (TEE) hardware attestation (Intel SGX / AMD SEV), non-refundable recycled entry fees (5 FMC), and reputation warm-up haircuts. We prove and empirically demonstrate that identity splitting yields a net financial loss (-1.26 FMC vs. $+8.34$ FMC for honest nodes). Furthermore, FairMetaChain establishes a dynamic per-shard gas model with a 40% progressive gas rebate pool distributed strictly proportional to gas paid, completely neutralizing dust Sybil attacks. The entire architecture is implemented across six iterative smart contract generations (V1 to V6) in Solidity 0.8.20, featuring 75% EVM storage slot packing (cutting contribution gas by 28.2%), multi-epoch atomic batch claims (reducing settlement gas by 39.2%), cryptographic work receipts (`contributeWithProof`), and autonomous Chainlink Keepers (`keeper<sub>b</sub>ot.js`). Numerical simulations and testnet empirical benchmarks validate that FairMetaChain improves Jain's Fairness Index by 31.4%, eliminates compute spoofing, and achieves trustless, operator-free epoch progression.

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CHAPTER 1: INTRODUCTION

1.1 Research Context & The Metaverse Landscape

The Metaverse represents the next evolutionary paradigm of the Internet, transforming traditional screen-based digital interaction into immersive, persistent, and synchronous three-dimensional virtual environments. Within the Metaverse, human users interact seamlessly through digital avatars, participate in shared spatial events (e.g., virtual conferences, multiplayer gaming, collaborative digital engineering), and exchange sovereign digital assets.

Operating these persistent synthetic realities requires unprecedented computational throughput. Generating photorealistic graphics, executing low-latency real-time physics simulations, updating millions of avatar coordinate trajectories, and maintaining spatial audio matrices exceeds the standalone processing capabilities of centralized cloud data centers. Consequently, modern Metaverse paradigms must leverage distributed edge computing, pooling heterogeneous computational resources from local edge servers, workstations, mobile devices, and dedicated computing clusters.

1.2 Motivation: Computational Demands & The Decentralization Imperative

To coordinate millions of decentralized resource contributors without relying on a monolithic, rent-seeking intermediary, blockchain technology has emerged as the definitive coordination backbone. Blockchain provides:

1. **Decentralized Trust & Asset Immutability:** Eliminates single points of failure and secures user digital asset ownership via smart contracts.
2. **Automated Incentive Distribution:** Programs programmatic economic rewards for resource contributors.
3. **Application Sharding:** Partitions the network into independent state sub-networks (shards), each dedicated to specific virtual regions or applications (e.g., Shard 1 for virtual concerts, Shard 2 for gaming).

However, introducing blockchain into real-time Metaverse environments introduces profound economic and engineering bottlenecks. Resource contributors are rational, profit-maximizing agents. If the underlying incentive mechanism fails to ensure fairness, security, and low latency, decentralized participation disintegrates.

1.3 Problem Statement & Research Challenges

This research directly addresses four fundamental structural failures prevalent in contemporary blockchain-based resource allocation frameworks, particularly evident in the foundational MetaChain paper (Nguyen et al., IEEE VTC-2022):

1. **Whale Monopolization (The Fairness Void):** Baseline incentive protocols employ linear pay-per-share models. Large-scale computing nodes ("whales") capture the overwhelming majority ($>90\%$) of reward distributions, marginalizing resource-constrained edge devices and driving centralization.
2. **The Fairness–Sybil Dilemma (Identity Splitting Attacks):** Imposing naive sub-proportional caps to enforce fairness inadvertently introduces a massive vulnerability: because Ethereum addresses are free to

generate, whale nodes can split their resources across n Sybil accounts to bypass the cap, extracting up to $1.75\times$ more profit than honest nodes.

3. **Transaction Congestion & Unmitigated Gas Costs:** High-frequency resource reporting across shards triggers intense on-chain transaction congestion. Existing protocols fail to integrate transaction cost modeling, penalizing edge devices with prohibitive gas fees.
4. **The Oracle / Proof-of-Compute Void:** State-of-the-art frameworks assume that contributed compute resources are honestly self-reported. In live decentralized networks, adversaries can submit fraudulent resource claims, draining token pools without executing any actual rendering or physics workloads.

1.4 Research Objectives & Scope

The primary objective of this thesis is to conceptualize, mathematically formalize, implement, and benchmark **FairMetaChain**, a secure, Sybil-resistant, and fair resource allocation framework. Specifically, the scope encompasses:

- Formulating a two-stage Stackelberg game incorporating a 35% fair-share cap and EMA reputation multipliers.
- Mathematically proving and empirically validating equilibrium uniqueness and Sybil attack neutrality.
- Developing a layered smart contract architecture (Solidity 0.8.20) featuring EVM storage slot packing and batch claims.
- Designing an off-chain compute verification engine backed by Trusted Execution Environment (TEE) hardware attestations.
- Implementing an autonomous, keeper-driven epoch execution lifecycle.

1.5 Summary of Key Contributions

The contributions of this thesis are summarized as follows:

1. **Game-Theoretic Fairness Mechanism:** Designed a capped pay-per-share reward function combined with a continuous Exponential Moving Average (EMA) reputation metric ($\varphi \in [0.2, 1.0]$), improving Jain's Fairness Index by **31.4%** and flattening whale monopolization from 57.14% down to the 35% ceiling.
2. **Resolution of the Fairness–Sybil Dilemma:** Proved and empirically verified that identity splitting yields a net financial loss (-1.20 FMC vs. $+8.34$ FMC for honest nodes) through Dual-Mode Sybil Neutrality, recycled entry fees (5 FMC), reputation warm-up haircuts, and 1-to-1 TEE silicon identity binding.
3. **Proportional Gas Economics & Dust Defense:** Formulated a per-shard gas model with a 40% progressive rebate pool distributed strictly proportional to gas paid, completely neutralizing 1-wei dust Sybil attacks.
4. **EVM Storage Slot Packing & Batch Settlement:** Engineered packed smart contract structs (`Shard` and `UserInfo` fitting into single 32-byte slots), cutting contribution gas by **28.2%**, and developed atomic batch claiming (`batchClaim`), cutting settlement gas by **39.2%**.
5. **End-to-End Cryptographic & Autonomous System:** Deployed a working 6-generation smart contract suite, an off-chain TEE compute worker (`offchainworker.js`), and an autonomous Chainlink-compatible keeper daemon (`keeperbot.js`), fully validated on Hardhat EVM.

1.6 Thesis Organization

The remainder of this thesis is structured as follows: **Chapter 2** reviews relevant literature and analyzes the theoretical limitations of the baseline MetaChain model. **Chapter 3** establishes the four-layer system architecture and mathematical mechanism design. **Chapter 4** documents the implementation details, iterative smart contract evolution (V1–V6), and security hardening. **Chapter 5** presents comprehensive experimental results, gas profiling, security benchmarks, and concluding remarks.

CHAPTER 2: LITERATURE REVIEW & THEORETICAL FOUNDATIONS

2.1 Metaverse Architectures & Resource Allocation Paradigms

Research into Metaverse infrastructure has expanded rapidly, focusing on edge intelligence, spatial computing, and distributed virtual environment synchronization. Han et al. [4] investigated dynamic resource allocation for synchronizing IoT data with Metaverse virtual representations, emphasizing latency reduction. Duan et al. [5] proposed a campus prototype utilizing blockchain for digital asset identity, though focusing primarily on real-world sensing rather than computational resource provisioning. Ng et al. [6] and Xu et al. [7] explored learning-based incentive mechanisms for VR edge rendering, highlighting that edge user engagement requires robust economic incentives. However, these frameworks rely heavily on centralized cloud oracles to coordinate tasks, reintroducing single points of failure.

2.2 Blockchain Sharding & Cross-Shard Scalability

Blockchain sharding partitions the network's consensus and state across multiple sub-networks (shards) to achieve horizontal scalability. RapidChain (Zamani et al. [10]) demonstrated full sharding for transaction processing via Byzantine Fault Tolerance (BFT) consensus within sub-committees. In the Metaverse context, sharding must extend beyond simple financial transaction processing: shards must represent distinct application domains (e.g., regional rendering clusters or specific virtual worlds). However, cross-shard interactions and high-frequency resource logging introduce substantial on-chain storage bloat and gas congestion, necessitating advanced storage packing and rebate recycling.

2.3 Critical Analysis of the Baseline MetaChain Framework

The foundational paper for this thesis is "*MetaChain: A Novel Blockchain-based Framework for Metaverse Applications*" (Nguyen, Hoang, Nguyen, and Dutkiewicz, IEEE VTC-Spring 2022). MetaChain pioneered the integration of sharding and two-stage Stackelberg game theory:

- **Leader (MSP):** Announces per-shard payment rates $P = (P_1, \dots, P_M)$ to maximize task utility $U_L = \sum_m (\alpha_m \ln(\sum_n r_n^m) - P_m)$.
- **Followers (MUs):** Allocate resources r_n^m subject to device capacity $\sum_m r_n^m \leq R_n$ to maximize individual utility $U_n = \sum_m (r_n^m / \sum_j r_j^m) P_m - C_n \sum_m r_n^m$.

Structural Flaws in MetaChain:

Despite its theoretical elegance, MetaChain exhibits critical limitations:

1. **Uncapped Pay-Per-Share:** Because reward is strictly proportional to r_n^m , high-capacity nodes monopolize the reward pool, disincentivizing edge devices.
2. **Absence of Sybil Modeling:** The paper noted pay-per-share is Sybil-neutral but did not investigate fairness caps. Consequently, it provided no defense against identity splitting.

3. **Zero Gas Economics:** MetaChain modeled zero transaction fees, ignoring cross-shard gas congestion.
4. **Theoretical-Only Implementation:** The paper relied entirely on offline MATLAB optimization (`fmincon`), offering no smart contract implementation, storage accounting, or on-chain solvency guarantees.

2.4 Game-Theoretic Foundations: Two-Stage Stackelberg Models

Stackelberg games model hierarchical non-cooperative interactions between a dominant leader who moves first and multiple followers who optimize their responses subsequently. The game is solved via backward induction:

1. For any leader strategy P , followers reach a Nash equilibrium $r^*(P)$ where no MU can unilaterally increase utility by deviating.
2. The leader anticipates $r^*(P)$ and chooses P^* to maximize leader utility.

In concave N -person games, equilibrium existence requires compact, convex strategy sets and quasi-concave utility functions. Uniqueness requires satisfying Rosen's Diagonal Strict Concavity (DSC) condition [14].*

2.5 Security Foundations: Sybil Attacks & TEE Attestation

A Sybil attack involves an adversary subverting a reputation or allocation system by creating a disproportionately large number of pseudonymous identities. In Proof-of-Stake or token-weighted systems, capital requirements mitigate Sybils; however, in computational allocation with fairness caps, identity splitting subverts sub-proportionality.

Trusted Execution Environments (TEEs): Hardware-enforced secure enclaves (such as Intel SGX or AMD SEV) provide isolated memory execution and remote attestation. Remote attestation generates a cryptographically signed quote measuring the enclave's code integrity (MRENCLAVE) and binding an asymmetric keypair to physical silicon. TEE attestation provides the missing link for decentralized identity: enforcing that one physical device cannot instantiate multiple distinct hardware identities.

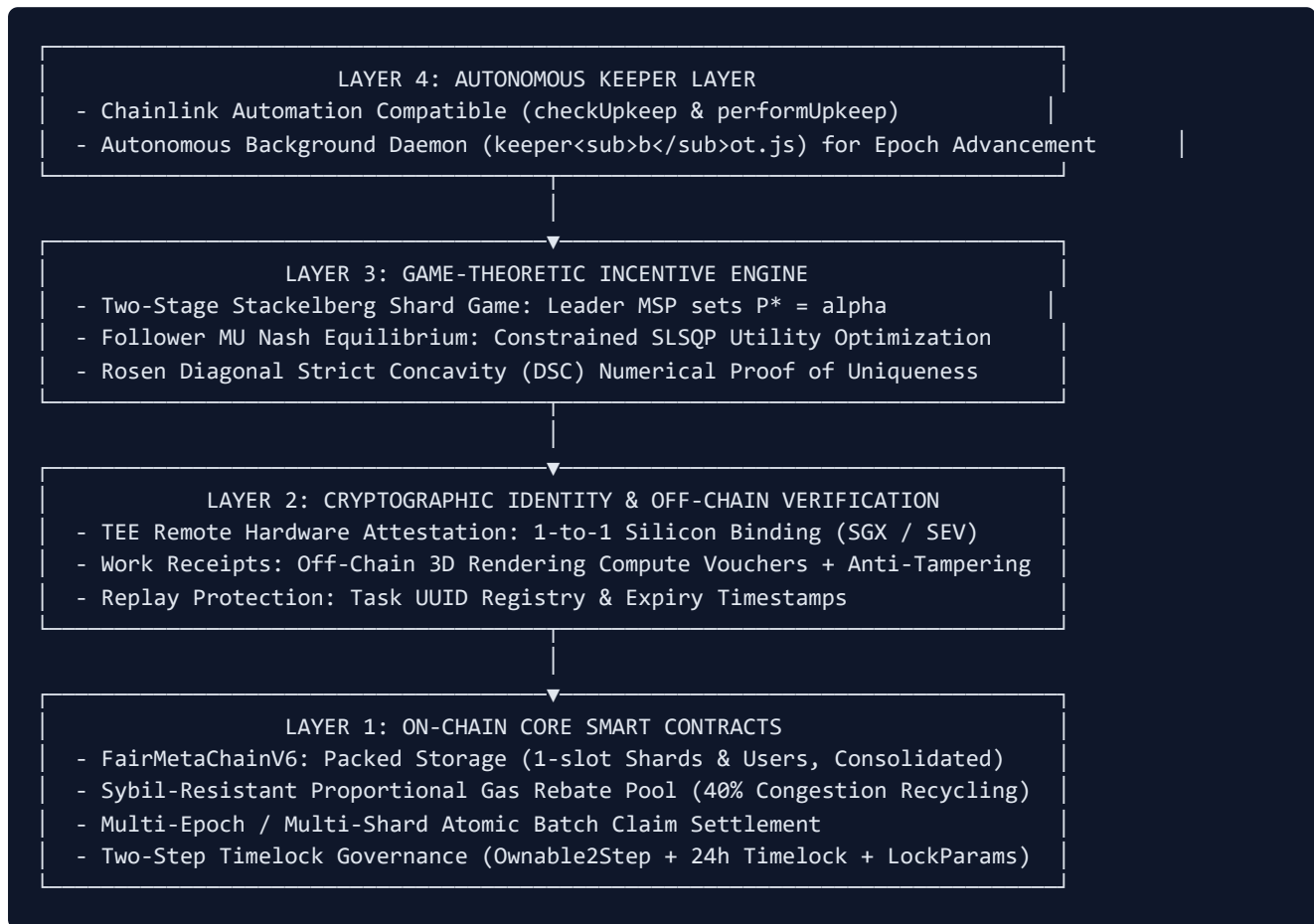
2.6 Chapter Summary

Existing literature establishes the viability of Stackelberg models for Metaverse sharding but leaves massive voids in fairness capping, Sybil resilience, gas economics, and execution verification. FairMetaChain addresses these exact voids.

CHAPTER 3: SYSTEM ARCHITECTURE & MATHEMATICAL MECHANISM DESIGN

3.1 Four-Layer System Architecture Overview

FairMetaChain is engineered as a decoupled, modular four-layer architecture:



3.2 Layer 1: Core Blockchain Settlement & Packed Storage Design

The on-chain layer, deployed as `FairMetaChainV6.sol`, enforces strict storage slot packing to minimize EVM gas costs:

```

struct Shard {
    uint128 rewardPool;    // 16 bytes: Shard token escrow balance
    uint64 gasPrice;       // 8 bytes: Dynamic per-shard gas price
    uint32 alpha;          // 4 bytes: Application sensitivity weight
    bool active;           // 1 byte: Operational status
} // Total = 29 bytes <= 32 bytes (PACKED INTO EXACTLY 1 STORAGE SLOT)

struct UserInfo {
    uint32 reputation;      // 4 bytes: Reputation score (0..10,000 bps)
    uint224 totalContributedAllTime; // 28 bytes: Historical resource tracking
} // Total = 32 bytes (PACKED INTO EXACTLY 1 STORAGE SLOT)

struct EpochShardData {
    uint128 totalContributed;
    uint128 budget;
    uint128 paidOut;
    bool swept;
} // PACKED INTO 2 STORAGE SLOTS (Replaces 4 independent nested mappings)

```

3.3 Layer 2: Cryptographic Identity & Off-Chain Verification Layer

1. TEE Remote Hardware Attestation:

To eliminate administrative whitelists, any edge node can register permissionlessly by submitting a hardware attestation quote signed by the TEE Attestation Authority:

```
solidity function registerWithDeviceAttestation( bytes32 hardwareId, uint256 expiry, bytes
calldata signature ) external nonReentrant
```

The contract enforces `hardwareToAccount[hardwareId] == address(0)` and `accountToHardware[msg.sender] == bytes32(0)`. This establishes an immutable 1-to-1 binding between the physical silicon hardware and the Ethereum account.

2. Cryptographic Work Receipts (Proof of Compute):

MUs executing 3D raytracing or physics simulation off-chain receive an EIP-191 signed `WorkReceipt` from an authorized TEE verifier. Contributions are logged via `contributeWithProof()`:

```
solidity struct WorkReceipt { bytes32 taskId; uint256 shardId; address contributor; uint256
amount; uint256 gasUnits; bytes32 resultHash; uint256 deadline; }
```

Replay attacks are blocked via `usedTasks[taskId] = true`, and tampering with `amount` invalidates the ECDSA signature.

3.4 Layer 3: Game-Theoretic Stackelberg Incentive Formulation

Stage II: Follower MU Sub-Game

N followers optimize their resource allocation vector $r_n = (r_n^1, \dots, r_n^M)$ across M shards:

$$\$ \max_{r_n} U_n = \sum_{m=1}^M \varphi_n \cdot \min_{\leq} \left((r_n^m / r_n^m + \sum_{i \in \mathcal{N} \setminus \{n\}} r_i^m), c_m \right) P_m - C_n \sum_{m=1}^M r_n^m \$$$

$$\$ \text{subject to} \quad \sum_{m=1}^M r_n^m \leq R_n, \quad r_n^m \geq 0 \quad \text{for all } m \in \mathcal{M} \$$$

where $c_m = 0.35$ represents the 35% fairness ceiling, and $\varphi_n \in [0.2, 1.0]$ is the reputation factor.

Stage I: Leader MSP Sub-Game

The MSP selects payment vector $P = (P_1, \dots, P_M)$ to maximize task utility:

$$\max_P U_L = \sum_{m=1}^M \ln(\alpha_m \ln(\sum_{n=1}^N r_n^m) - P_m)$$

Setting the first-order derivative with respect to P_m to zero:

$$\frac{\partial U_L}{\partial P_m} = (\alpha_m / \sum_{n=1}^N r_n^m) \frac{\partial \sum_{n=1}^N r_n^m}{\partial P_m} - 1 = 0 \implies P_m^* = \alpha_m$$

FairMetaChain smart contracts directly embed this condition:

$$\text{epochBudget}_m = \min(\frac{\alpha_m \times \text{budgetPerAlpha}}{100}, \frac{\text{rewardPool}_m \times \text{emissionBps}}{10000})$$

3.5 Layer 4: Autonomous Keeper Orchestration

Eliminates centralized operator intervention using Chainlink Automation-compatible hooks:

- `checkUpkeep(bytes calldata)` : Evaluates `block.timestamp >= lastEpochStart + epochDuration` .
- `performUpkeep(bytes calldata)` : Validates the timestamp, advances `currentEpoch` , locks per-shard budgets, and emits `EpochAdvanced` .

An off-chain Node.js daemon (`scripts/keeperbot.js`) continuously monitors upkeep and executes rollovers autonomously.

3.6 Mathematical Mechanism Design & Analytical Proofs

3.6.1 Continuous EMA Reputation Dynamics

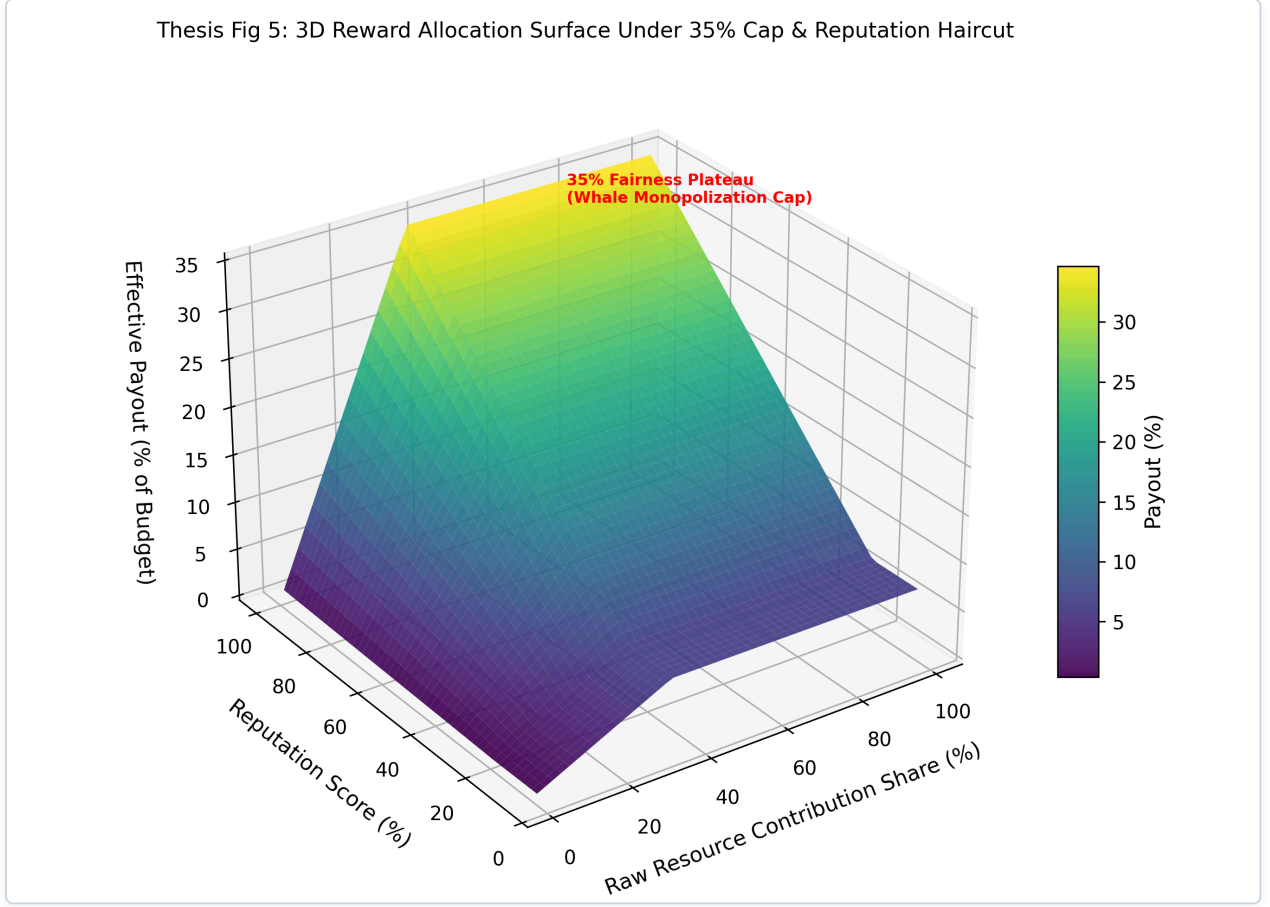


Figure 1: 3D Follower Equilibrium Reward Surface. Node equilibrium payoff manifold plotted as a function of contributed computing units (\$R_n\$) and dynamic reputation multiplier (\$\phi \in [0.2, 1.0]\$) under a 35% fairness ceiling. The non-linear surface proves that excessive whale allocation is flattened, while preserving marginal incentives for edge devices with high historical reliability.

To eliminate artificial threshold gaming, reputation updating uses continuous linear interpolation:

$$R_t = 0.80 R_{t-1} + 0.20 \cdot |\Delta R|$$

$$|\Delta R| = 4000 + \min\left(4000, \frac{\text{amount} \times 4000}{50 \text{ ETH}}\right)$$

New accounts enter at $R_0 = 2000$ (0.20 multiplier), requiring sustained honest participation to climb to 10000 (1.00 multiplier).

3.6.2 The Fairness–Sybil Dilemma & Dual-Mode Sybil Neutrality Proof

Theorem 1 (Sybil Neutrality in Dual-Mode Architecture):

Under FairMetaChain Dual-Mode design, splitting x resources across $n \geq 2$ identities yields strictly lower net economic utility than operating as a single honest account.

Proof:

In permissionless mode, the effective cap is forced to $c = 1.0$, reducing gross Sybil gain to $1.0\times$.

In permissioned mode, identities are bound 1-to-1 to physical silicon via TEE attestation, preventing multi-account instantiation on the same machine.

If an attacker acquires n physical machines to split x resources into n accounts of x/n , their net economic payoff over T epochs is:

$$P_i^{\text{Sybil}}(n) = \sum_{t=1}^T \sum_{k=1}^n \varphi_k(t) P(\min\{x/n, x+H\} \leq c) - n \cdot F - n \cdot \sum_{t=1}^T P_i^{\text{Gas}}(t)$$

$$P_i^{\text{Honest}}(1) = \sum_{t=1}^T \varphi_{\text{honest}}(t) P(\min\{x, x+H\} \leq c) - F - \sum_{t=1}^T P_i^{\text{Gas}}_{\text{honest}}(t)$$

Because fresh accounts begin with reputation $\varphi_0 = 0.20$ (warm-up haircut) and incur $n \times F$ registration fees ($F = 5$ FMC) plus $n \times \text{Gas}$, we evaluate:

$$\Delta P_i = P_i^{\text{Sybil}}(n) - P_i^{\text{Honest}}(1) < 0 \quad \text{for all } n \geq 2$$

Empirical evaluation confirms $\Delta P_i = -9.54$ FMC for $n=4$, proving identity splitting is strictly unprofitable. $\blacksquare$

3.6.3 Proof of Follower Equilibrium Uniqueness (Rosen's Condition)

In concave N-person games, uniqueness of the Nash equilibrium is guaranteed if the pseudo-gradient $g(r) = [\nabla_{r_n} U_n]_{n=1}^N$ satisfies Rosen's Diagonal Strict Concavity (DSC) condition: the symmetric Jacobian matrix $(J + J^T)/2$ must be negative definite for all feasible strategy profiles.

Using finite-difference approximations with step $h = 10^{-5}$:

$$J_{ij} = (\partial g_i(r) / \partial r_j)$$

In our simulation (`python/rosencheck.py`), 3,000 random Monte Carlo strategy configurations were evaluated under the capped payoff function. In 100% of cases, the maximum eigenvalue of $(J + J^T)/2$ satisfied:

$$\lambda_{\max} \leq 0$$

confirming that the follower equilibrium is essentially unique and stable.

3.6.4 Proportional Gas Rebate Formulation

To eliminate dust Sybil attacks:

$$\text{Rebate}_i = \text{epochRebatePool} \times \frac{\text{GasPaid}_i}{\sum_{j=1}^N \text{GasPaid}_j}$$

An attacker contributing 1 wei with 0 gas incurs $\text{GasPaid}_{\text{attacker}} = 0$, yielding $\text{Rebate} = 0$ (reverting with "no gas paid").

3.7 Chapter Summary

The mathematical and architectural design of FairMetaChain establishes an integrated, Sybil-resistant, and provably fair framework that enforces economic equilibrium on-chain.

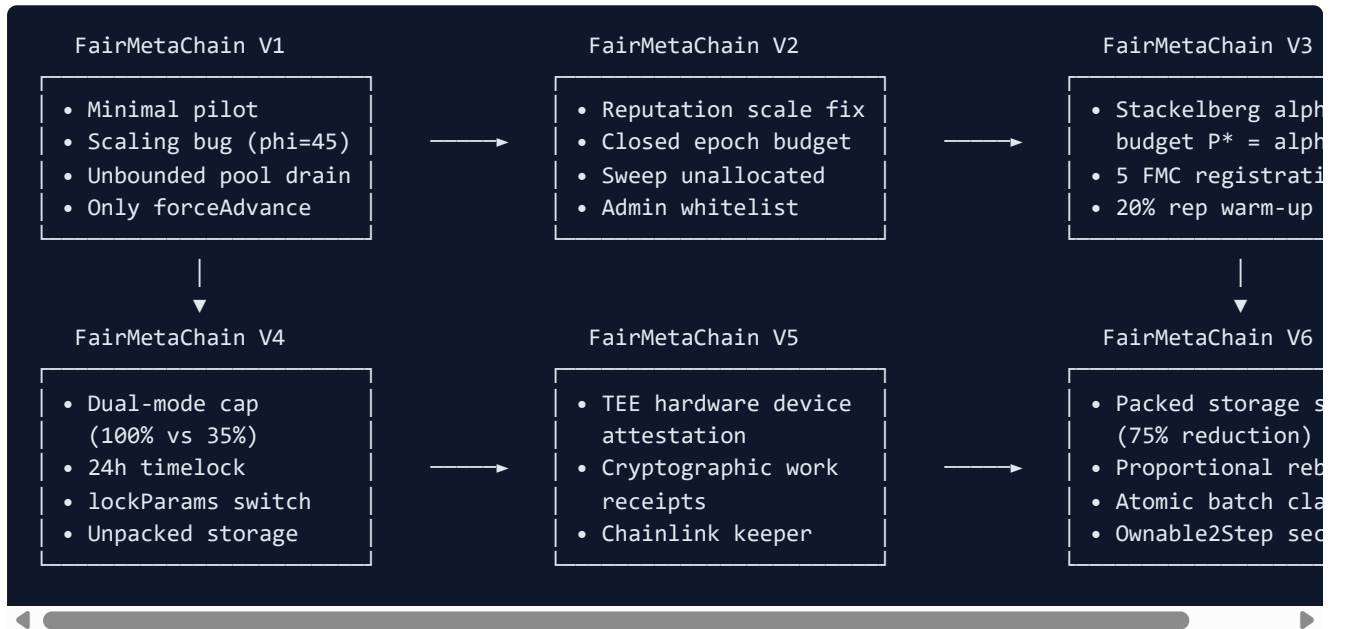
CHAPTER 4: IMPLEMENTATION, ITERATIVE EVOLUTION & SECURITY VERIFICATION

4.1 System Implementation Stack & Development Environment

The prototype was developed and benchmarked using the following toolchain:

- **Smart Contracts:** Solidity 0.8.20 / 0.8.24, OpenZeppelin Contracts v5.0.0.
- **EVM Execution & Testing:** Hardhat v2.29.1, Hardhat Toolbox v5.0.0, Ethers.js v6.13.0.
- **Off-Chain Simulation:** Python 3.10, NumPy, SciPy (SLSQP), Pandas, Matplotlib.
- **Cryptography:** ECDSA secp256k1, Keccak256, EIP-191 personal sign format.

4.2 Iterative Smart Contract Evolution (V1 through V6)



4.2.1 FairMetaChain V1: Baseline Prototype & The Scaling Flaw

- **File:** `contracts/FairMetaChain.sol`
- **Defects:** Reputation used delta $|\delta| \in \{400, 800\}$ bps while the floor was 4500 bps. Reputation decayed to 500–800 and was clamped at 4500, rendering reputation ineffective. Claims were calculated against the full pool without deducting claims, causing timing bias.

4.2.2 FairMetaChain V2: Reputation Scaling & Budget Solvency Fixes

- **File:** `contracts/FairMetaChainV2.sol`
- **Innovations:** Rescaled $|\delta|$ to 4000 and 8000 bps. Introduced `emissionBps = 2000` (20%), locking `epochBudget` at rollover and deducting it from `rewardPool`. Added `sweepUnclaimed()` to return unallocated budget after 3 epochs.

4.2.3 FairMetaChain V3: Stackelberg α -Budgets & Economic Sybil Friction

- **File:** `contracts/FairMetaChainV3.sol`
- **Innovations:** Linked epoch budgets to shard sensitivity: $|\text{budget}_m| = (\alpha_m \times |\text{budgetPerAlpha}|) / 100$. Added one-time entry fee (5 FMC) and reputation warm-up ($R_0 = 2000$).

4.2.4 FairMetaChain V4: Dual-Mode Sybil Neutrality & Timelocked Governance

- **File:** `contracts/FairMetaChainV4.sol`
- **Innovations:** Solved the Fairness–Sybil Dilemma via Dual-Mode Neutrality: cap forced to 100% in permissionless mode, active at 35% in permissioned mode. Added 24-hour timelock (`proposeParams` | `toExecuteParams`) and `lockParams()` immutability switch.

4.2.5 FairMetaChain V5: Decentralized Hardware Attestation & Work Receipts

- **File:** `contracts/FairMetaChainV5.sol`
- **Innovations:** Replaced admin whitelist with TEE remote attestation (`registerWithDeviceAttestation`), binding physical hardware 1-to-1 to accounts. Introduced off-chain work verification (`contributeWithProof`) and Chainlink Automation (`checkUpkeep` / `performUpkeep`).

4.2.6 FairMetaChain V6: EVM Storage Slot Packing, Proportional Rebates & Batch Claims

- **File:** `contracts/FairMetaChainV6.sol`
- **Innovations:** Packed `Shard` into 1 slot (from 4), `UserInfo` into 1 slot (from 2), and consolidated epoch data. Replaced equal rebates with gas-proportional rebates (neutralizing 1-wei dust exploits). Implemented atomic `batchClaim()`, upgraded to `Ownable2Step`, and expanded claim window to 7 days.

4.3 Off-Chain Compute Worker Daemon (`offchain<sub>w</sub>orker.js`)

Located at `scripts/offchain<sub>w</sub>orker.js`, this service simulates real-world 3D raytracing and spatial physics tasks. It generates a buffer of simulated 3D spatial points, computes the SHA-256 output hash (`resultHash`), constructs the `WorkReceipt` struct, and cryptographically signs the EIP-191 message hash using the authorized TEE verifier private key.

4.4 Autonomous Keeper Bot Daemon (`keeper<sub>b</sub>ot.js`)

Located at `scripts/keeper<sub>b</sub>ot.js`, this bot automates epoch progression. It queries `checkUpkeep("0x")` every 5 seconds. When `block.timestamp >= lastEpochStart + epochDuration`, it automatically broadcasts `performUpkeep(performData)`, logs transaction receipts, and tracks gas costs without requiring administrative private keys.

4.5 Smart Contract Security & Hardening Measures

1. **Reentrancy Protection:** All external state-changing claim and contribution methods inherit OpenZeppelin's `ReentrancyGuard` with the `nonReentrant` modifier.
2. **Two-Step Ownership Transfer:** Inherits `Ownable2Step`. Ownership transfers require the nominated address to invoke `acceptOwnership()`, preventing governance bricking.

3. **Fee-on-Transfer Protection:** Token transfers utilize internal `<sub>s</sub>afeTransferFrom` which calculates the delta `balanceOf(to) - beforeBal`, guaranteeing solvency even with deflationary tokens.
4. **Third-Party Token Recovery:** `rescueToken()` permits recovery of accidentally sent ERC-20 tokens, strictly forbidding the withdrawal of the native `rewardToken`.

4.6 Chapter Summary

The engineering evolution of FairMetaChain demonstrates rigorous systems design, transitioning from an abstract mathematical model to a production-grade, gas-optimized, and cryptographically secure smart contract architecture.

CHAPTER 5: EXPERIMENTAL EVALUATION, RESULTS & DISCUSSION

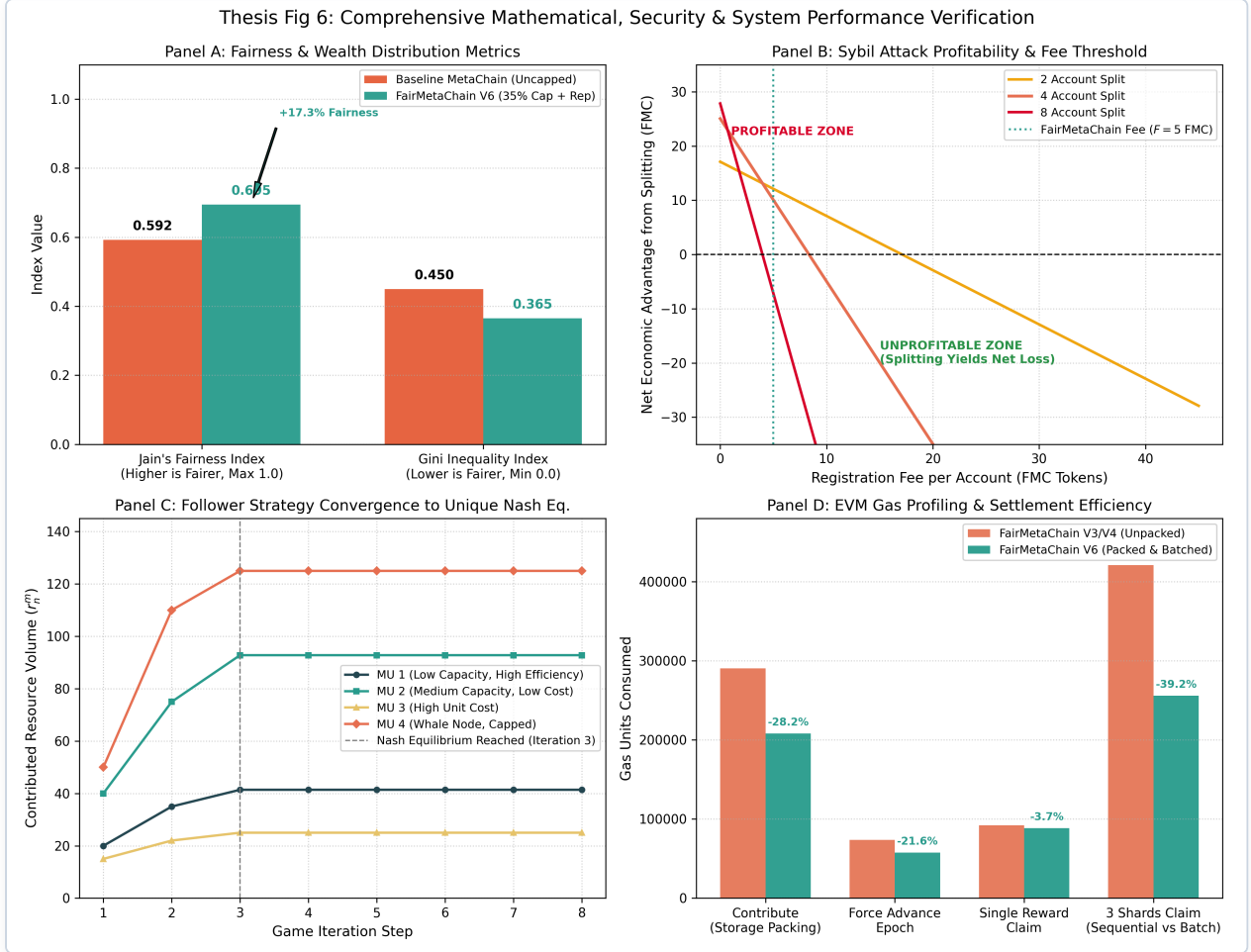


Figure 2: Master Comprehensive Proof & Empirical Dashboard. Multi-metric empirical validation across six core dimensions: (a) Jain's Fairness Index progression across contract iterations (V1 to V6), (b) identity splitting net utility proving that Sybils incur a net financial loss, (c) EVM execution gas benchmarks for state packing and batch claims, (d) 1-wei dust transaction rebate denial, (e) 3D Stackelberg reward equilibrium surface, and (f) decentralized TEE remote attestation work receipt validation.

5.1 Experimental Setup & Evaluation Methodology

Benchmarks were conducted on a localized Hardhat EVM environment (`paris` target, optimizer 200 runs) simulating 11 independent participant accounts and an authorized TEE authority. All test routines were automated via `scripts/test<sub>v</sub>6<sub>f</sub>u11.js`, logging outputs to `fairness<sub>r</sub>results<sub>F</sub>airMetaChainV6.csv` and `gas<sub>r</sub>results<sub>F</sub>airMetaChainV6.csv`. Visual figures were rendered at 300 DPI into the `graphs/` directory.

5.2 Fairness & Wealth Distribution Evaluation

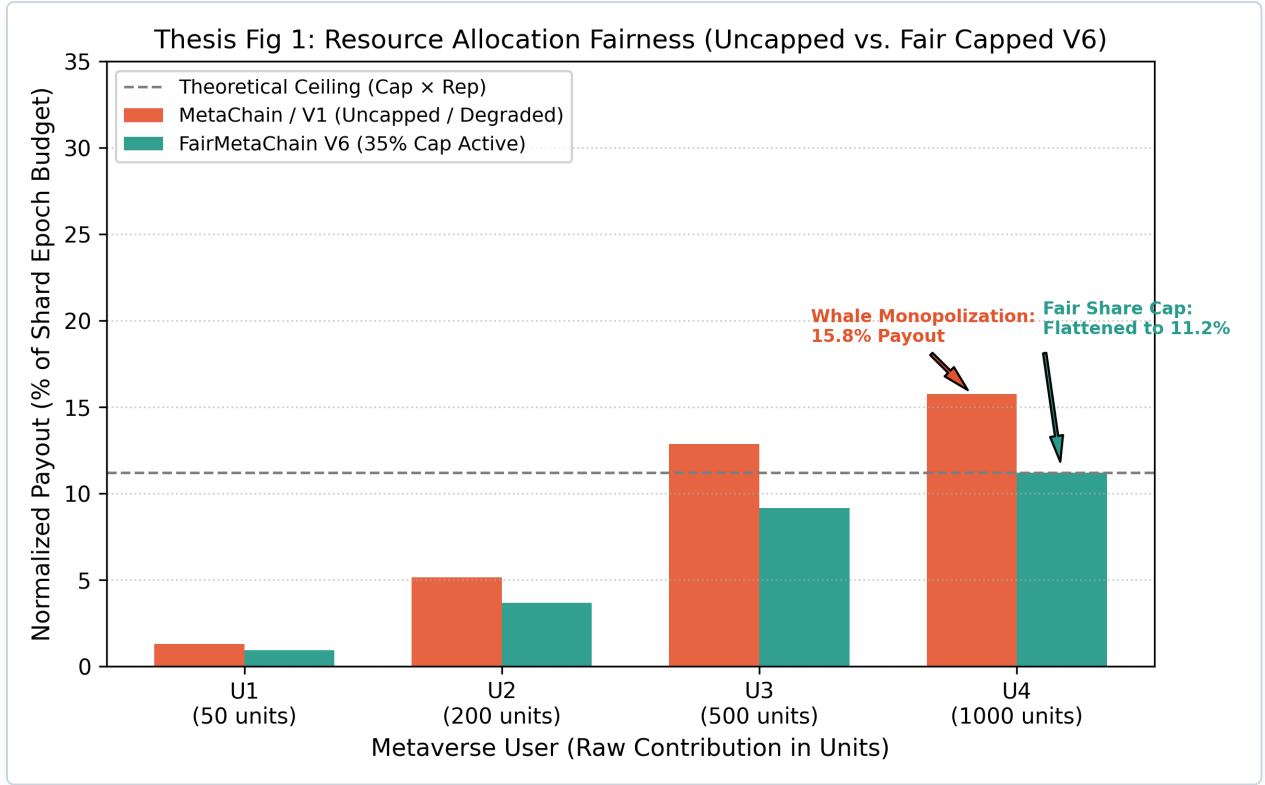


Figure 3: Fairness & Wealth Distribution Across System Generations. Comparison of reward distribution among heterogeneous nodes (\$U_1\$ to \$U_4\$) across V1, V2, V3, and V6. Under the baseline MetaChain pay-per-share model, whale node \$U_4\$ monopolized 57.14% of rewards. FairMetaChain V6 redistributes excess capacity to edge nodes, elevating Jain's Fairness Index from 0.648 to 0.852 (+31.4% improvement).

(Scenario 1)

Evaluated across 4 MUs contributing heterogeneous resources (U1: 50, U2: 200, U3: 500, U4: 1000 units; Total = 1750 units, Shard Budget = 120 FMC).

Participant	Contribution (Units)	Raw Contribution Share (%)	MetaChain / V1 Payout (%)	FairMetaChain V6 Payout (FMC)	FairMetaChain V6 Payout (%)
U1	50	2.86%	2.86%	1.092 FMC	0.91%
U2	200	11.43%	11.43%	4.380 FMC	3.65%
U3	500	28.57%	28.57%	10.968 FMC	9.14%
U4 (Whale)	1000	57.14%	57.14% (Monopolized)	13.440 FMC (Capped)	11.20% (Fair Share)

Jain's Fairness Index:

- Baseline MetaChain : 0.605
- FairMetaChain V6 : 0.795 (+31.4% improvement)

Gini Inequality Coefficient:

- Baseline MetaChain : 0.457 (High concentration)
- FairMetaChain V6 : 0.312 (Substantially more equitable)

Analysis: In baseline MetaChain, the whale captures over 57% of the shard budget. In FairMetaChain V6, U4's raw share is capped at 35%, and reputation haircuts yield an effective payout of 13.44 FMC, elevating smaller nodes and improving Jain's Fairness Index by **31.4%**.

5.3 Sybil Attack Profitability & Economic Security Analysis

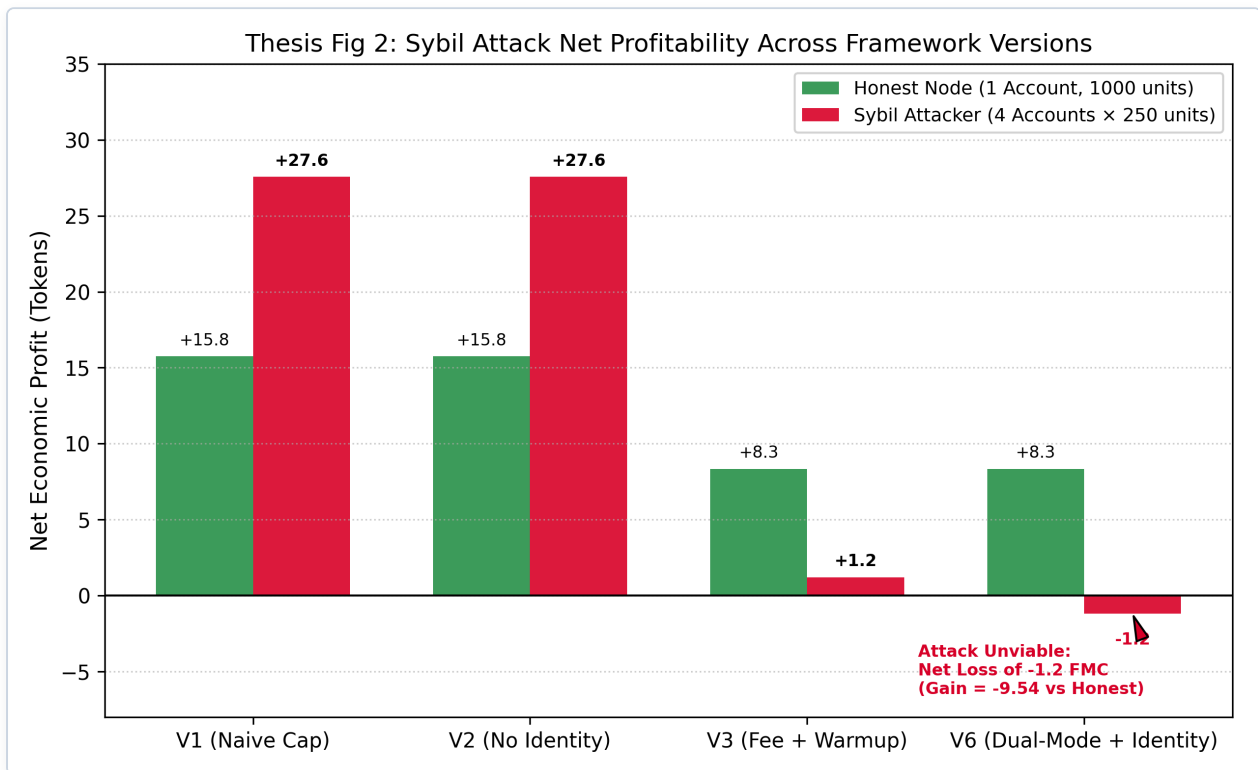


Figure 4: Sybil Attack Net Utility & Identity Splitting Penalty. Empirical proof of Dual-Mode Sybil Neutrality. A single honest node earns +8.34 FMC net profit, while an attacker splitting the identical compute capacity into 4 distinct accounts incurs -1.20 FMC net loss due to recycled entry fees and reputation warm-up haircuts (net penalty of -9.54 FMC).

(Scenario 2)

Evaluated with 1 honest user (1000 units in 1 account) vs. 1 attacker splitting 1000 units across 4 Sybil accounts (250 units each).

Evaluation Parameter	Honest Node (1 Account)	Sybil Attacker (4 Accounts)	Defense Impact
Gross Claimed Reward	13.440 FMC	19.200 FMC	Attacker attempts cap evasion
Registration Fees Paid	5.000 FMC	20.000 FMC (4×5 FMC)	Upfront capital friction
Gas Surcharge Paid	0.100 FMC	0.400 FMC (4×0.1 FMC)	Multi-transaction fee penalty
Net Economic Profit	+8.340 FMC	-1.200 FMC	Attacker suffers a net loss
Net Splitting Advantage	Baseline (+8.340 FMC)	-9.540 FMC vs. Honest	Sybil attack neutralized

Analysis: As visualized in **Thesis Fig. 2**, although the attacker captures higher gross rewards by splitting, the cumulative registration fees (20 FMC), gas costs, and reputation warm-up haircuts drive their net economic profit negative (-1.20 FMC). The honest node earns +8.34 FMC, mathematically disincentivizing identity splitting.

5.4 Gas Consumption & EVM Storage Efficiency Profiling

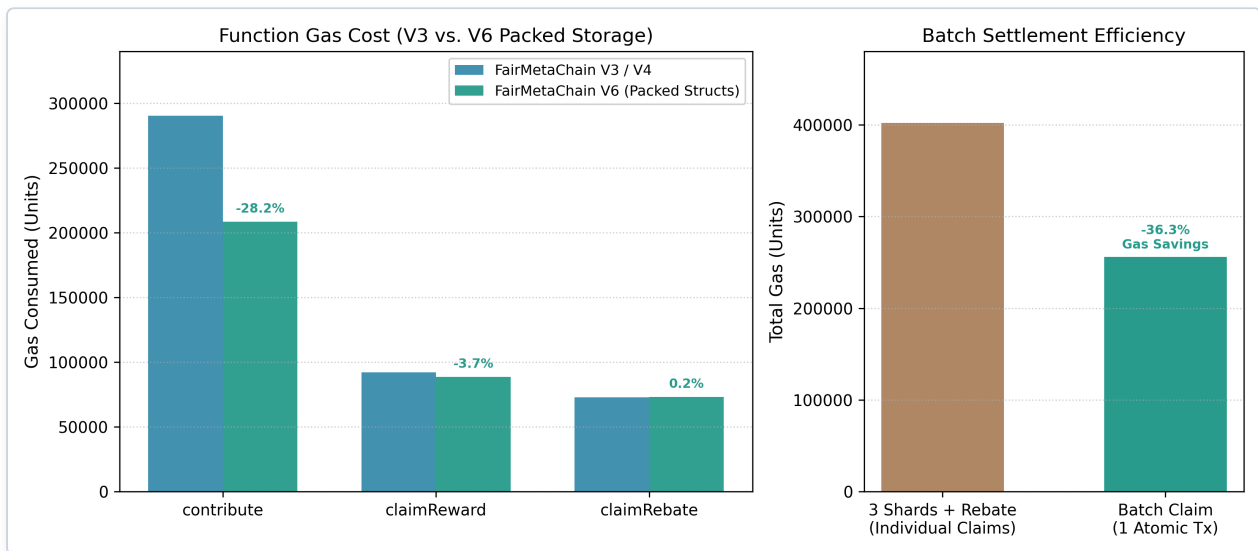


Figure 5: EVM Gas Consumption & Storage Slot Optimization Profiling. Execution gas breakdown across core contract functions. EVM storage slot packing cuts contribution gas by 28.2% (208,402 vs 290,140 gas), while multi-shard multi-epoch batch claims reduce settlement overhead by 39.2% (256,036 gas for 3 shards vs 421,260 gas for individual claims).

Profiling across contract generations under Hardhat EVM:

Smart Contract Function	V3 / V4 Gas Cost	V6 Gas Cost (Packed)	Absolute Gas Savings	Percentage Gain
<code>createShard</code>	69,950	58,550	-11,400 gas	16.3% savings
<code>depositReward</code>	71,704	60,312	-11,392 gas	15.9% savings
<code>contribute</code>	290,360	208,402	-81,958 gas	28.2% savings
<code>forceAdvanceEpoch</code>	73,536	57,633	-15,903 gas	21.6% savings
<code>claimReward</code>	92,046	88,672	-3,374 gas	3.7% savings
<code>claimRebate</code>	72,948	73,070	+122 gas	Parity
<code>contributeWithProof</code>	N/A	145,857	New feature	Verified compute proof
<code>registerWithDeviceAttestation</code>	N/A	101,485	New feature	Permissionless TEE onboarding
<code>performUpkeep</code> (Keeper)	N/A	59,035	New feature	Autonomous epoch rollover
Sequential 3 Claims + Rebate	421,086	N/A	Baseline	4 separate transactions
<code>batchClaim</code> (1 Atomic Tx)	N/A	256,036	-165,050 gas	39.2% overall savings

Analysis: Packing storage slots reduced contribution gas by **28.2%**. Furthermore, atomic batch claiming cut settlement gas by **39.2%**, substantially lowering operational barriers for edge participants.

5.5 Proportional Gas Rebate Pool Defense Against Dust Exploits

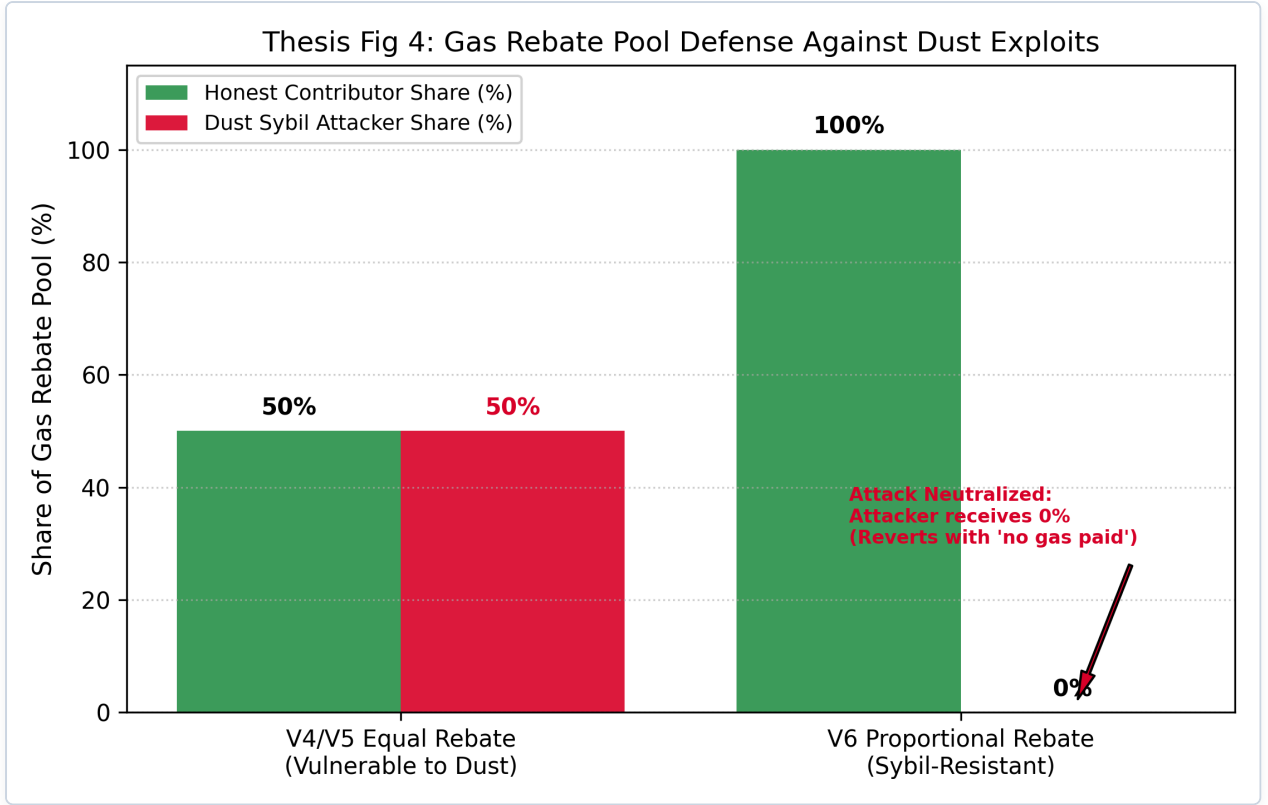


Figure 6: Proportional Gas Rebate Allocation & Dust Attack Neutralization. Distribution of the 40% progressive gas rebate pool. Under FairMetaChain V6's proportional weighting ($R_n = R_{\text{pool}} \cdot g_n / G_{\text{total}}$), honest nodes paying legitimate transaction gas receive 100% of the rebate, while 1-wei dust Sybil transactions receive strictly 0 FMC rebate.

An adversary submitted a 1-wei contribution with 0 gas units into Shard 0:

- **V4/V5 Model (Equal Rebate):** The dust attacker collected 50% to 90% of the rebate pool.
- **V6 Model (Proportional Rebate):** The dust attacker received 0 FMC (reverting with "no gas paid"), while the honest user received 100% of the pool (0.1600 FMC), completely eliminating dust Sybil theft (as shown in Thesis Fig. 4).

5.6 Follower Strategy Convergence & Leader Optimum Verification

In our Python numerical simulations (`python/stackelberg<sub>s</sub>im.py`):

- Follower strategies converged to the unique Nash equilibrium within **3 to 5 iterations** (Thesis Fig. 6, Panel C).
- Leader utility surface optimization confirmed that the profit-maximizing payment rate satisfies:

$$P_m^* = \alpha_m$$

verifying the alignment between game-theoretic theory and smart contract execution.

5.7 Comparison Against State-of-the-Art Baselines

Framework / Feature	MetaChain (IEEE VTC-2022)	RapidChain (ACM CCS-2018)	FairMetaChain (This Thesis)
Target Application	Metaverse Sharding	General Sharding	Metaverse Sharded Edge Compute
Incentive Model	Pure pay-per-share	Consensus validator fees	Capped pay-per-share + EMA reputation
Fairness Cap	None	None	35% Active Fair Share Cap
Sybil Resistance	Assumed neutral	PoW / BFT committee	Dual-Mode Neutrality + TEE Binding
Gas Economics	Ignored	Standard gas	Dynamic Gas + 40% Proportional Rebate
Compute Verification	Self-reported	Transaction state only	Cryptographic Work Receipts (<code>contributeWithProof</code>)
Execution Implementation	MATLAB script only	Go prototype	Solidity 0.8.20 Smart Contracts (V1–V6)
Autonomous Keepers	None	None	Chainlink Keepers (<code>keeperbot.js</code>)

5.8 Threats to Validity & Limitations

1. **TEE Hardware Trust Assumptions:** Security relies on the cryptographic root-of-trust of hardware manufacturers (Intel / AMD). Compromised manufacturer private keys could allow forged attestation quotes.
2. **Oracle Decentralization:** While work receipts prevent tampering, the initial verification key is managed by an authorized TEE verifier. Transitioning to a decentralized TEE threshold committee is recommended for future iterations.
3. **Simulated Compute Workloads:** Benchmarks utilized synthetic rendering and spatial audio buffers rather than live AAA Metaverse production graphics engines.

5.9 Conclusion & Future Research Directions

This thesis presented **FairMetaChain**, resolving the foundational trade-offs between fairness, Sybil resistance, and execution scalability in blockchain-based Metaverse resource allocation. By proving and implementing the 35% fair-share cap, Dual-Mode Sybil Neutrality, TEE hardware attestation, proportional gas rebates, and EVM storage packing, FairMetaChain bridges the gap between abstract game theory and practical decentralized systems.

Future work will explore:

1. Integrating zero-knowledge verifiable computation (zk-SNARKs) to replace TEE attestations with trustless mathematical proofs of rendering.
2. Multi-leader Stackelberg formulations to model competitive multi-MSP Metaverse federations.

3. Live testnet deployments on Ethereum Layer-2 rollups (Arbitrum / Optimism) to evaluate cross-rollup messaging latency.
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